Solutions to Linear Algebra Problems

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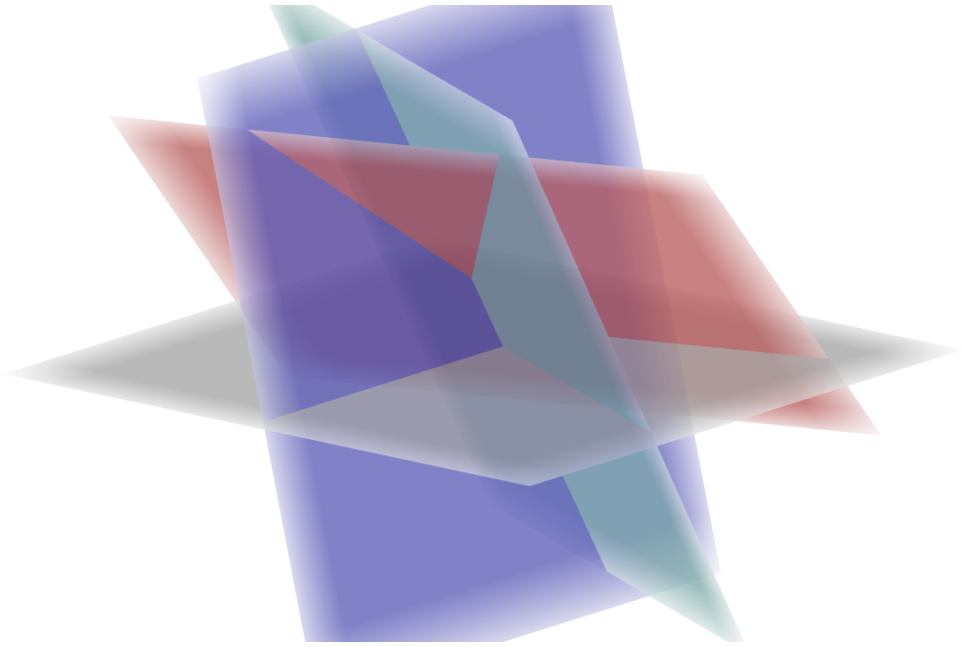
Abstract

This note collects my solutions to selected problems in Linear Algebra. The goal is to present clear arguments and organized solutions for study and review. Books that I use for assignments sources include:

- Linear Algebra Done Right 4e, Sheldon Axler



Problem 1.

Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Source: Linear Algebra Done Right - Ex.12 p.25

Solution:

Let V_1, V_2 be two subspaces of V.
($\Rightarrow$): Suppose that $V_1 \cup V_2$ is a subspace of V. Suppose for contradiction that one of the subspaces V_1 or V_2 is not contained in the other. We can choose u, w such that $u \in V_1, u \notin V_2$ and $w \in V_2, w \notin V_1$.
Since $V_1 \cup V_2$ is a subspace, we have $u + w \in V_1 \cup V_2$. This means $u + w$ must be in V_1 or V_2.
If $u + w \in V_1$, since V_1 is a subspace and $-u \in V_1$, then $w = (u + w) - u \in V_1$, which is a contradiction. Similarly, if $u+w \in V_2$, since V_2 is a subspace and $-w \in V_2$, then $u = (u+w)-w \in V_2$, which is also a contradiction.
Therefore, $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$.
($\Leftarrow$): If $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$, then $V_1 \cup V_2 = V_2$ or $V_1 \cup V_2 = V_1$, which is a subspace.

Problem 2.

Suppose $v_1, \ldots, v_m$ is a list of vectors in V. For $k \in \{1, \ldots, m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that the list $v_1, \ldots, v_m$ is linearly independent if and only if the list $w_1, \ldots, w_m$ is linearly independent.

Source: Linear Algebra Done Right - Ex.14 p.38

Solution:

($\Rightarrow$): Suppose the list $v_1, \ldots, v_m$ is linearly independent. Consider a combination of $w_1, \ldots, w_m$ such that:

$$\begin{aligned} c_1w_1 + c_2w_2 + \cdots + c_mw_m &= 0 \\ \Rightarrow c_1v_1 + c_2(v_1 + v_2) + \cdots + c_m(v_1 + v_2 + \cdots + v_m) &= 0 \\ \Rightarrow (c_1 + c_2 + \cdots + c_m)v_1 + (c_2 + \cdots + c_m)v_2 + \cdots + c_mv_m &= 0 \end{aligned}$$

Since $v_1, \ldots, v_m$ are linearly independent, it follows that $c_1 = c_2 = \cdots = c_m = 0$. Therefore, the list $w_1, \ldots, w_m$ are linearly independent.
($\Leftarrow$): Suppose $w_1, \ldots, w_m$ are linearly independent. Consider a combination of $v_1, \ldots, v_m$ such

that:

$$\begin{aligned} c_1v_1 + c_2v_2 + \cdots + c_mv_m &= 0 \\ \Rightarrow c_1w_1 + c_2(w_2 - w_1) + c_3(w_3 - w_2) + \cdots + c_m(w_m - w_{m-1}) &= 0 \\ \Rightarrow (c_1 - c_2)w_1 + (c_2 - c_3)w_2 + \cdots + (c_{m-1} - c_m)w_{m-1} + c_mw_m &= 0 \end{aligned}$$

Since $w_1, \ldots, w_m$ are linearly independent, it follows that $c_1 = c_2 = \cdots = c_m = 0$. Therefore, the list $v_1, \ldots, v_m$ are linearly independent.

Problem 3.

Suppose V is finite-dimensional and U, W are subspaces of V such that $V = U + W$. Prove that there exists a basis of V consisting of vectors in $U \cup W$.

Source: Linear Algebra Done Right - Ex.5 p.43

Solution:

Let $u_1, u_2, \ldots, u_m$ be a basis of U and $w_1, w_2, \ldots, w_n$ be a basis of W.
Since $V = U + W$, we have the list $\{u_1, u_2, \ldots, u_m, w_1, w_2, \ldots, w_n\}$ spans V.
Applying the procedure in Theorem 2.30 to reduce this list to a basis of V produces a basis consisting of the vectors $u_1, \ldots, u_m$ and some of the w's (none of the u's get deleted in this procedure because $u_1, \ldots, u_m$ is linearly independent).